The thesis – sound propagation in ducts with absorbent linings

My thesis at the Acoustics Laboratory, from September 1988 to January 1989, was about silencers: how sound propagates in a duct whose walls are lined with a porous, sound-absorbing material. The task was to set up a calculation model from the literature, program it and test it against measurements. My supervisor was Finn Jacobsen. Here is a summary – the full thesis and the interactive recalculations are also on the site.

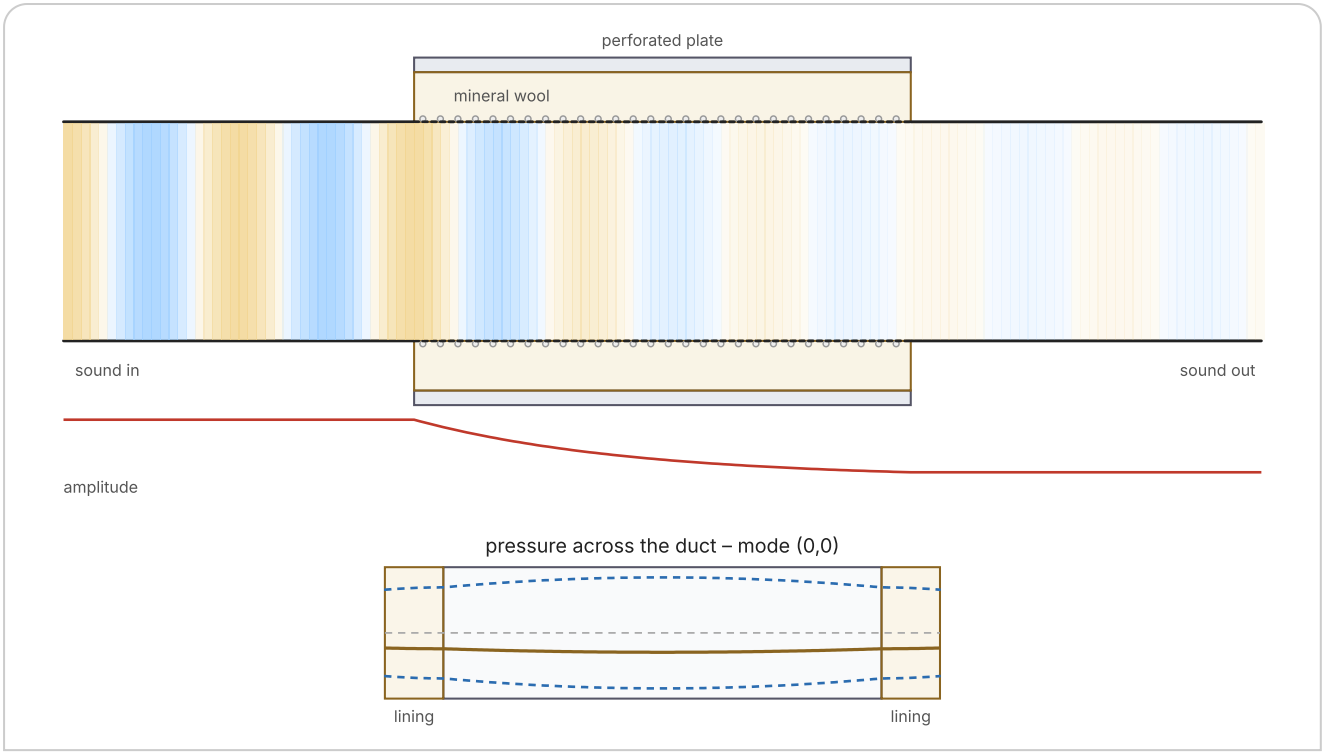
> Thesis, Acoustics Laboratory, DTH (1989), 36 points: *Sound Propagation in Ducts with Absorbent Linings* (Danish title: Lydtransmission i absorptionsbeklædte rør). Supervisor: Finn Jacobsen. The thesis: PDF · Figure 3.10(b) recomputed interactively · the HP BASIC program "Extended" translated to JavaScript · full duct model

The problem

Many noise problems are caused by sound carried through pipes and ducts: ventilation systems, chimneys and exhaust systems. The sound is reduced with silencers. In ventilation, *absorptive silencers* are used above all, where the duct walls are lined with mineral wool behind a perforated plate, so that sound is converted into heat while the air flows freely. In a hard duct sound travels almost without loss over long distances (see standing waves in tubes); with an absorbent lining the propagation constant Γ acquires a real part, and the sound pressure decays exponentially with distance:

$$p(x,y,t) = \psi(y)\, e^{-\Gamma x} e^{j\omega t}, \qquad \Gamma = \alpha + j\beta, \qquad D = 8.686\,\alpha \ \ \mathrm{dB/m}$$

How much does a given lining attenuate at a given frequency? It is not an easy question, for the sound field in the duct is not a plane wave, and sound also propagates inside the lining itself.



Morse and Scott

Philip Morse treated the problem in 1939 assuming *local reaction*: the lining is described by a surface impedance, and the sound at a point on the surface depends only on the pressure right there. That is reasonable if the material has a high flow resistance or is partitioned by closely spaced baffles, so that sound cannot run along the lining. But for ordinary mineral wool it goes wrong – Morse's theory predicts, for instance, that the attenuation tends to infinity as the air gap becomes narrow.

In 1946 R. A. Scott developed a theory of *extended reaction* (bulk reaction): sound propagates both in the air gap and in the lining, and the two fields share the same propagation constant Γ. Continuity of pressure and normal particle velocity at the interface and a rigid outer wall give a transcendental equation for the allowed modes. D. R. A. Christie wrote it in 1971 in dimensionless form:

$$f(w) = w\tan w + M\sqrt{w^2+p^2}\,\tan\!\left(b\sqrt{w^2+p^2}\right) = 0$$

where $w = \sqrt{(\Gamma\ell)^2 + (k\ell)^2}$ is the transverse wavenumber in the air gap of half-width ℓ, $p = \sqrt{(h\ell)^2 - (k\ell)^2}$ is set by the wavenumber h in the lining, $b = d/\ell$ is the ratio of the lining thickness to the half-width, and M is the ratio of the complex densities of air and lining. Once w is found, $\Gamma = \sqrt{(w/\ell)^2 - k^2}$. The equation has infinitely many complex roots – one for each mode – and must be solved numerically.

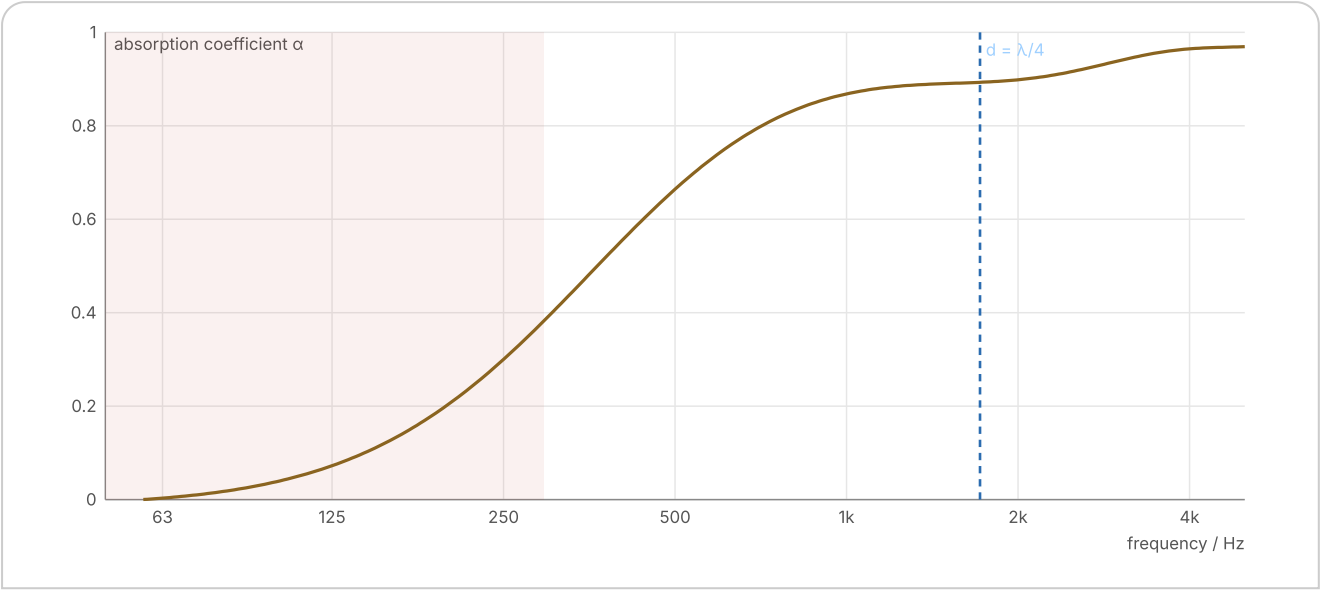
The lining

To use Scott's equation one must know the wavenumber and characteristic impedance of the lining. In 1970 M. E. Delany and E. N. Bazley measured a large number of fibrous materials and found that both quantities depend, to a good approximation, on a single parameter: the ratio of frequency to the material's flow resistivity σ, $X = \rho_0 f/\sigma$:

$$Z_c = \rho_0 c\left[1 + 0.0571 X^{-0.754} - j\,0.087 X^{-0.732}\right], \quad k_c = \frac{\omega}{c}\left[1 + 0.0978 X^{-0.700} - j\,0.189\right.$$

The formulas hold for $0.01 < X < 1$. F. P. Mechel extended them in 1976 to low frequencies, and the thesis routine "Material" uses Mechel's version. For a lining of thickness d on a rigid wall the surface impedance is $Z_s = -jZ_c\cot(k_c d)$, and the normal-incidence absorption coefficient follows from the reflection coefficient.

Interactive absorption



The numerics

Scott's equation is solved frequency by frequency. At low frequencies the roots are approximately known: for the plane mode (0,0) $w \approx 0$, and for the first symmetric transverse mode (0,2) $w \approx \pi$. From these starting values each root is followed upwards in frequency, the previous solution – linearly extrapolated – serving as the guess for the next. The iteration itself uses Newton–Raphson's method:

$$w_{n+1} = w_n - \frac{f(w_n)}{f'(w_n)}$$

Newton–Raphson converges quickly when the guess is good, but can “jump” to another mode when the roots lie close together. The program “Extended” therefore compares the length of the solution vector from step to step and, on a jump, switches to Muller’s method, which approximates the function by a parabola through three points and is more robust for complex roots. The program was written in HP BASIC; the printouts are reproduced in § 3.5 of the thesis, and a translation to JavaScript can be run here on the site.

The anomalous modes

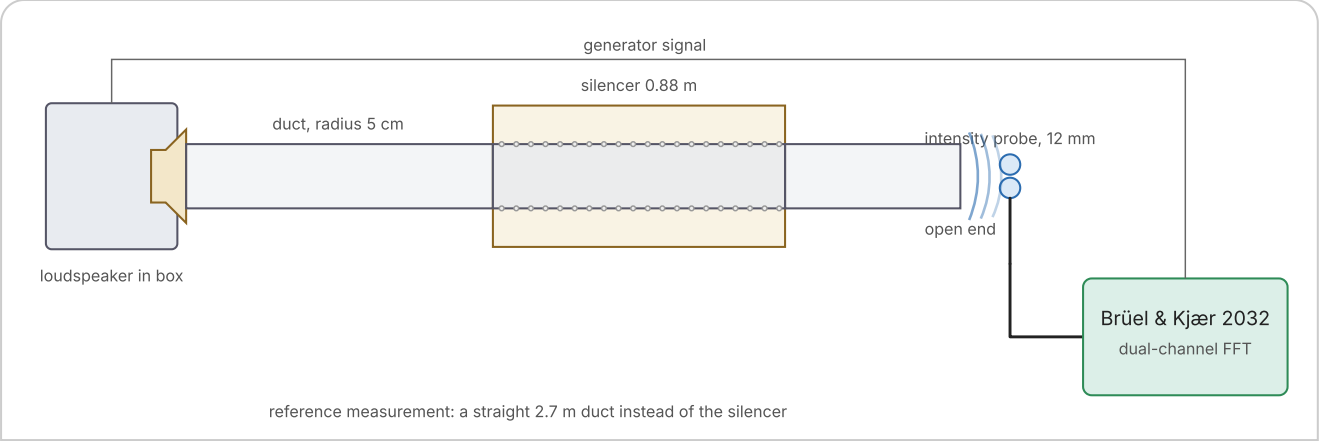
In a hard duct it is straightforward: the plane wave, mode (0,0), propagates at all frequencies, and the higher modes can propagate only above their cut-on frequency. The plane mode is always the least attenuated, so it determines the effect of the silencer far down the duct.

The most surprising result of the study was that this does not always hold in a lined duct. Above the cut-on frequency of the (0,2) mode, the (0,0) mode can be attenuated *more* than the (0,2) mode – for certain ratios of lining thickness to duct width and certain flow resistivities. It is as if the sound energy is being diffracted out towards the sides of the duct, so that the wavefront is no longer plane. The idea of separate modes, so clear in a hard duct, is probably not fully applicable in a soft one. Figure 3.10(b) of the thesis, which shows this for $\sigma = 8500$ N·s/m⁴ and $d = 0.05$ m, can be recomputed interactively.

The measurements

The algorithm was tested against published calculations and measurements (Christie 1971, Wassilieff 1987) and against my own measurements on a circular silencer kindly provided by Ødegaard & Danneskiold-Samsøe A/S: length 0.88 m, duct radius 5 cm and outer radius of the lining 10 cm. I measured the silencer’s *insertion loss* – the difference in radiated sound power with and without the silencer, relative to a straight reference duct – with a sound-intensity probe with a 12 mm spacer and a Brüel & Kjær 2032 dual-channel FFT analyser up to 3.2 kHz (see signal analysis). Measurements were made both with and without the perforated cylinder in front of the lining.

The flow resistivity of the lining was determined from dynamic measurements of flow resistance and lay between 30 000 and 40 000 N·s/m⁴. Calculations with “Extended” for the two limiting values agreed far better with the measurements than Morse’s theory. The remaining discrepancies could mainly be attributed to the reactive sound field at the ends of the silencer and the flexibility of the duct walls, which Scott’s theory does not take into account.



The conclusions

- The (0,0) and (0,2) modes behave anomalously above the cut-on frequency of (0,2): the plane mode is not always the least attenuated.
- Below cut-on for (0,2), the (0,0) mode is always the least attenuated. There is thus at least one frequency region in which sound propagation in lined ducts can be calculated safely.
- Agreement with published results is very good; Figure 3.10 agrees excellently with Wassilieff's calculations and measurements.
- The algorithm could be improved by a better definition of a "jump" and by calculating two modes simultaneously to obtain a composite curve of least attenuation – which the interactive recalculation on the site now does.

The thesis ends with a quotation from Niels Steensen (Nicolas Steno), anatomist, geologist and later bishop, beatified in 1988 – the year the work was written: *"Beautiful is what we see, more beautiful what we understand, most beautiful by far what we do not comprehend."*

Since 1989

- **Numerical methods:** Where Scott's equation then required careful root-finding, silencers are now computed with the finite element method in full three dimensions, with flow, perforations and end effects.
- **Modern mode tracking:** The problem of following roots that lie close together and swap places is well known; today the argument principle of complex analysis (contour integrals) is used to find all roots in a region – the duct model on the site shows a Nyquist plot of exactly this kind.
- **Better material models:** Delany–Bazley is still used, but Miki (1990) corrected the model's physical inconsistency at low frequencies, and the Johnson–Champoux–Allard model (1987–91) describes porous materials from five physical parameters.
- **Active noise control in ducts** with microphone, signal processor and loudspeaker has since the 1990s been able to reduce the low frequencies where absorbers are too thin.
- **Acoustic metamaterials** with resonators smaller than the wavelength promise thin silencers for low frequencies.

Further reading

- P. M. Morse: “The transmission of sound inside pipes”, *J. Acoust. Soc. Am.* 11 (1939) 205–210
- R. A. Scott: “The propagation of sound between walls of porous material”, *Proc. Phys. Soc.* 58 (1946) 358–368
- D. R. A. Christie: “Theoretical attenuation of sound in a lined duct”, *J. Sound Vib.* 17 (1971) 283–286
- M. E. Delany and E. N. Bazley: “Acoustical properties of fibrous absorbent materials”, *Applied Acoustics* 3 (1970) 105–116
- M. L. Munjal: *Acoustics of Ducts and Mufflers* (2nd ed., Wiley 2014)

- On this site: Sound in ducts with flow · Sound attenuation in ducts · The silencer in 3D · From Bell’s inequality to sound intensity

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